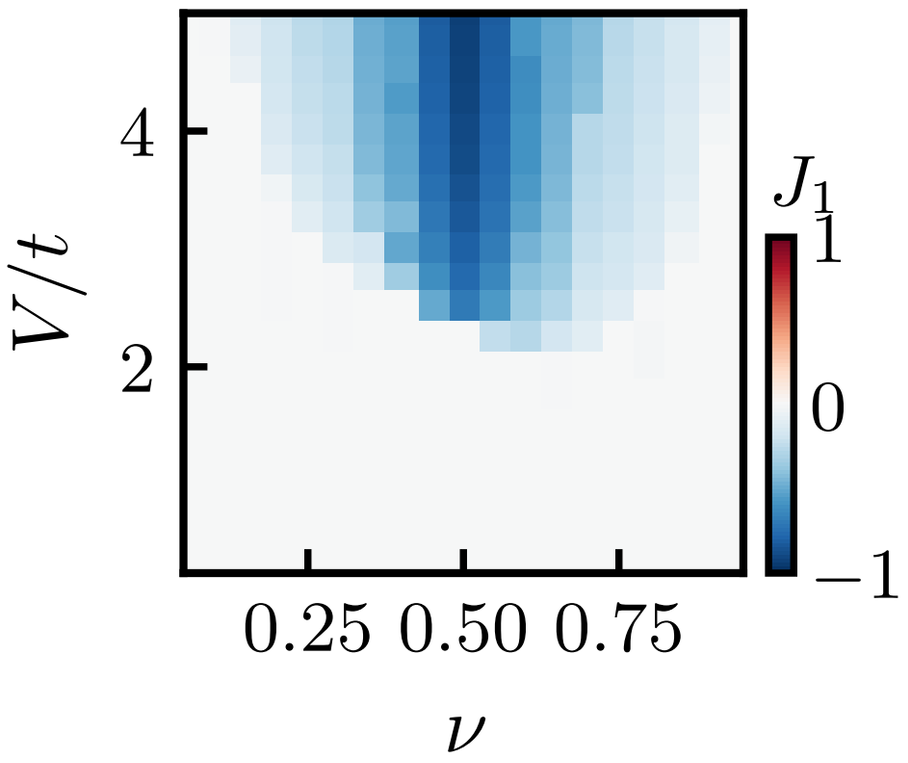
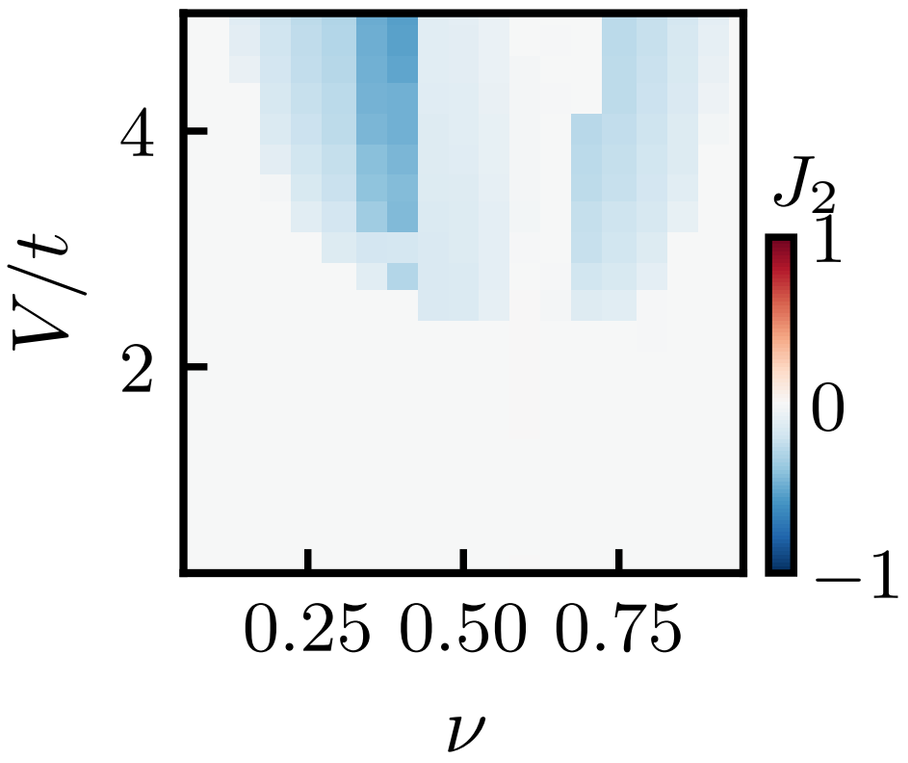
$\Phi = 0$

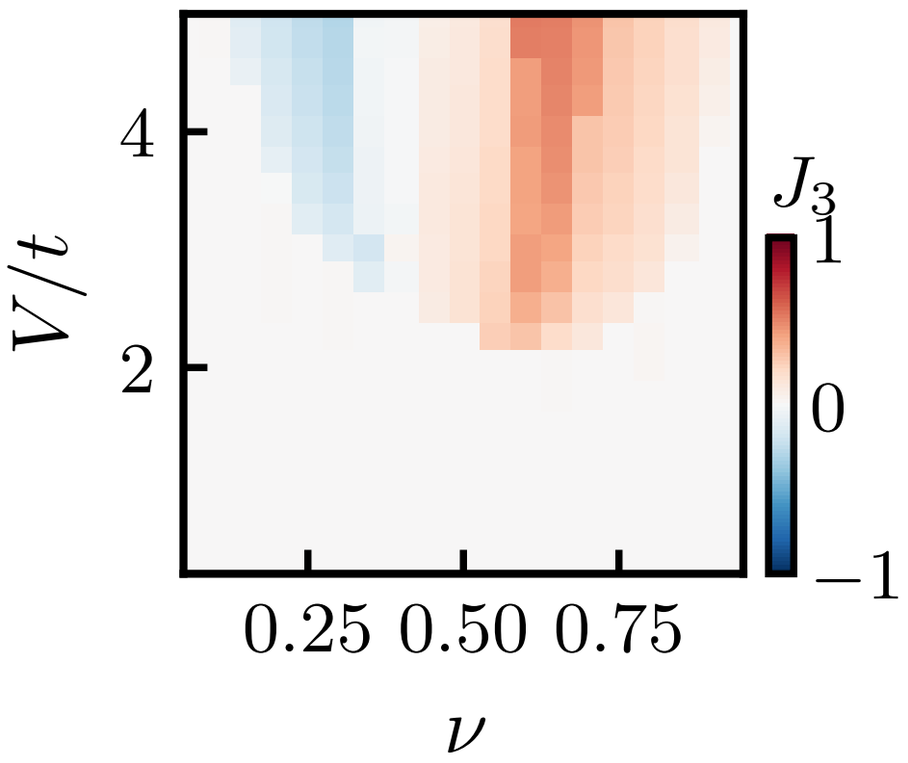
(a)


4
2
V/t
0.25 0.50 0.75
ν
J_1
1
0
−1

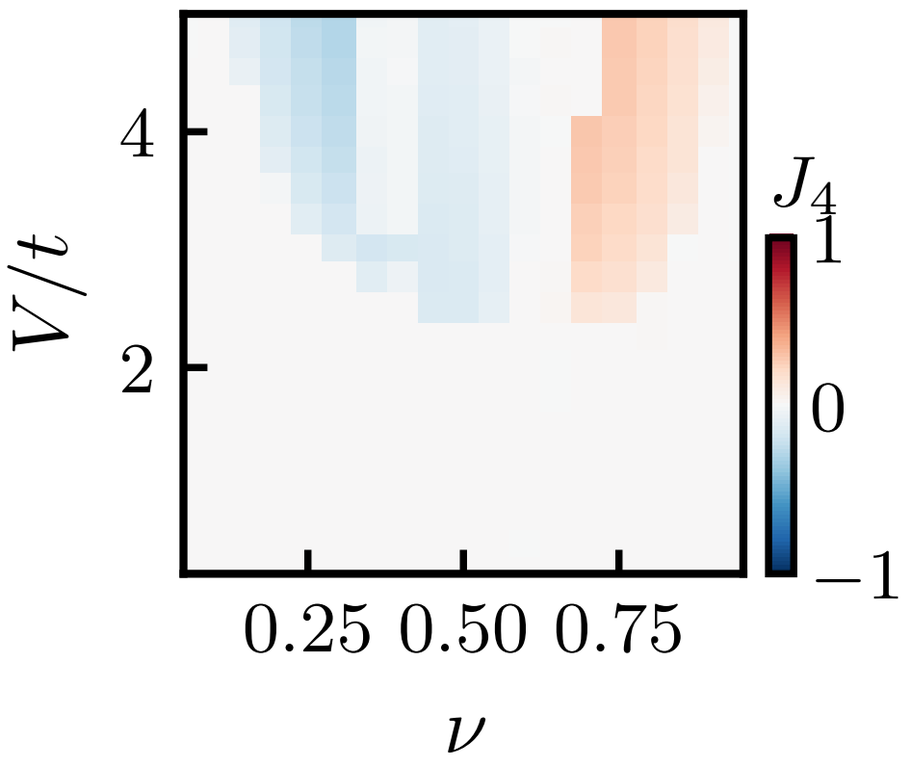
(b)


4
2
V/t
0.25 0.50 0.75
ν
J_2
1
0
−1

(c)


4
2
V/t
0.25 0.50 0.75
ν
J_3
1
0
−1

(d)


4
2
V/t
0.25 0.50 0.75
ν
J_4
1
0
−1